

FROM REGISTRATION TO REGULATION: UNCOVERING SMS SCAM PREVALENCE AND PATTERNS AFTER THE SIM CARD REGISTRATION ACT USING GENERALIZED ADDITIVE MIXED MODELS

Allanna Joy B. Tapang¹, Randel P. Oropel², Neil Angelo B. Gautane³, and Alleah Nicole D. Quimpo⁴

¹College of Science, Bulacan State University (Main Campus)

Abstract

The Philippines has experienced high cases of SMS scams, which were further intensified during the pandemic. To address this, the government implemented the SIM Card Registration Act. However, numerous reports indicated the continued occurrence of scam attacks despite its enactment. This study aimed to determine the prevalence and patterns of SMS scams after the implementation of the law using Generalized Additive Mixed Models. SMS data from BulSU students (2023–2025) were analyzed, with 3,267 (3.20%) of 102,311 messages identified as scams. Findings showed that SMS scams persisted due to the policy's limited scope, particularly in addressing smishing, which comprised 57.1% of cases. Scammers were found to route messages through unauthorized or unidentified cell towers and frequently impersonate money-related institutions. Clear temporal patterns were observed, with scam activity rising as early as 5 AM, peaking at 11 AM, and remaining elevated until around 8 PM. Scam activity was also associated with increases during key events, particularly volcanic activity and post-typhoon periods (approximately 50%). Additionally, although not part of the initially identified events, scam activity was associated with an approximate 80–83% decline following the crackdown on POGO operations in 2024, followed by a rebound in 2025 as scammers adapted their strategies. These findings indicated that SMS scam activity remained adaptive and responsive to real-world conditions. Lastly, the study provided possible policy improvements, suggested network security measures, and promoted actions toward public awareness against SMS scams.

Keywords: *GAMM, Generalized Additive Mixed Model, SIM Card Registration, Smishing, SMS Scam, Spoofing*

INTRODUCTION

Before the COVID-19 pandemic, the Philippines was already experiencing high cases of online fraud. A 2019 national household survey showed that one out of four Filipinos (24%) has been victims of online scams with SMS scams being the most frequently reported cybercrime incident (Albert et al., 2021). The emergence of the pandemic has further intensified this situation. As people relied heavily on their mobile phones and online transactions, scammers found more chances to attack.

From March to September 2020, SMS scams rose by 37% compared to the same months in 2019 (Cudis, 2021). The Philippine Central Bank also recorded a 2,324% increase in hacking cases

and a rise of 302% in phishing in comparison to 2019 (Hilotin, 2022). Due to this sudden surge in text scams, lawmakers introduced a policy that would reduce scams and identify offenders.

In October 2022, President Ferdinand Marcos Jr. signed RA No. 11934, also known as the SIM Card Registration Act, which has the goal of mitigating the widespread existence of SMS scams all over the Philippines (Parrocha, 2022). The law requires all users to register their SIM cards under their legal identity.

During the first several months after the implementation, the National Telecommunications Commission (NTC) reported a 93.3% drop in scam complaints (Parrocha, 2023). However, many Filipinos still continued to receive scam messages despite the enactment of the law. From a global perspective, countries such as Mexico repealed their SIM registration law after three years, as it was found to be ineffective and offers more risks than advantages (Yang, 2022). In 2024, a study by Abraham et al. (2024) of the Global Anti-Scam Alliance reported that Filipinos lost about 460 billion pesos in a year due to scams, most of which came from SMS, offering too-good-to-be-true deals such as job opportunities and financial assistance. These findings show that the problem remains strong, and that the current policy needs further review and improvement.

Existing studies about SMS scams solely focus on the detection of the scam and the perception and attitudes of the people with regards to the law. Hence, this study aimed to address these gaps by examining the policy's effectiveness in reducing scam messages by determining the prevalence and pattern of scamming to further improve policy, network security, and public awareness. Moreover, this research supported the United Nations Sustainable Development Goal (SDG) 16 that promotes peace, justice, and strong institutions by contributing to safer digital environments, offering insights that can improve policy, and strengthening protection against online fraud and other cyber-related crimes.

Objectives of the Study

This study aims to uncover the prevalence and patterns of SMS scams among students at Bulacan State University after the enactment of the SIM Registration Act of 2022. Specifically, this study aims to:

1. uncover the frequent origin of SMS scams in terms of telecommunication networks and the entities most scams are impersonating.
2. analyze the difference in distribution of SMS scams received by different telecommunication networks.
3. evaluate the temporal patterns of SMS scams in terms of the hour of day, day of week, and month of year.
4. analyze the contribution of different events (e.g. disasters, payday) to the increase of SMS scams.
5. recommend possible policy improvements, public awareness, and fortification of network security.

Related Theories

Three related theories form the basis for a comprehension of SMS scam dynamics: Routine Activity theory, Crime Pattern theory, and Opportunity theory. Routine Activity theory points to three elements necessary for crime to occur: (1) a motivated offender (scammer), (2) a suitable target

(students), and (3) the absence of a capable guardian (government). If any of these were missing, then crime was less likely to occur (Cohen & Felson, 1979). Aside from these, Crime Pattern theory puts forward the idea that the targets' whereabouts and schedule are also contributors to crime occurrence — suggesting crimes do not happen randomly but follow identifiable patterns (Brantingham & Brantingham, 2021). Furthermore, Opportunity theory suggested that another reason for these identifiable patterns were favorable opportunities that scammers take advantage of (Felson & Clarke, 1998). These theories highlighted that scam dynamics were shaped by time, context, and human behavior.

Review of Related Literature

Going back to ancient civilizations, Egypt, Greece, Rome, and China already had cases of scams and fraudulent acts, deceiving merchants to counterfeit currency (der Veen, 2023). Now with the advancements on technology, Bangko Sentral ng Pilipinas (BSP) identified different types of physical and online scamming, where phishing scams had become a lot more prevalent during the pandemic, especially smishing. The Philippine landscape has reached an alarming level of scams showing that 67% of Filipinos had received scams, translating to a total of 459.98 billion pesos worth of financial loss (Abraham et al., 2024). The given statistics show that the government was calling for a fight against the increasing number of scams in the Philippines.

To address these rising scam events, the government implemented the SIM Card Registration Act in 2022 with an aim to provide law enforcement against malicious use of mobile devices. The law requires all users to register their SIM cards within a given time frame and addresses the intention of combating spoofing. While the policy strengthened authentication and monitoring, it remained limited in its scope, as other types of scamming such as smishing were not fully addressed. Lastly, Telco networks and their respective users were expected to act in accordance with their own responsibilities. However, the implementation of the Mobile Portability Act allowed users to retain their existing numbers when switching to another Telco network. As a result, scam messages linked to a mobile number can continue even if the user changes network providers, instead of stopping when the service provider was unchanged.

Smishing and spoofing were common forms of online fraud that relied on deception and impersonation to target individuals. Spoofing is a type of cybercrime activity where someone pretends to be a trusted source to gain access to an individual's personal information or steal money, which can be in the form of calls, messages, websites, and URLs (Lenaerts-Bergmans, 2025). On the other hand, smishing is executed through deceptive content within the body of the message itself (Yasar, 2023). For further distinction between the two, messages originating from random and unknown numbers that rely solely on the message content for deception were considered as smishing, while a message that uses the name of a trusted company or entity as the sender's name constitutes spoofing. This distinction was important since only spoofing was defined in RA 11934; hence, it was not within the scope of the act to protect users from SMS scams that came from random numbers.

In the Philippine context, these two scamming techniques dominated the SMS scam landscape. Embudo (2022) and Dela Cruz (2025) agreed that there has been a continuous rise in scam messages and malicious links, many of which were associated with online gambling, loans, and fraudulent

services. Despite regulatory efforts, Rosales (2025) reported that gambling-related scam links continued to increase significantly. Similarly, Respicio & Co. (2025) and Gonzales (2024) noted that financial institutions are frequent targets, with scams attempting to obtain sensitive information such as OTPs and account details. Meanwhile, Dela Cruz (2024), GLOBE (2024b), GLOBE (2024a), and Guiao (2024) emphasized that although millions of scam messages have been blocked, scammers still bypass security measures through spoofed sender IDs and illegal cell towers. These findings highlighted the persistent and evolving nature of SMS scams and the limitations of existing regulations in fully protecting users.

Review of Related Studies

The current digital environment has seen an alarming rise in SMS scams, which evolved into a persistent and adaptive global threat. SMS remained the most dominant channel for scams because of its wide reach and trust in Telco networks (Abraham et al., 2024). This growing complexity underscored the severity of SMS scams as a persistent and adaptive threat in the current digital environment.

For a solution, the SIM Card Registration Act represented a major policy intervention aimed at combatting these scam crimes. However, despite its implementation, scams had remained prevalent due to significant loopholes and lack of comprehensive scope (Tumang, 2025). Noticing that perpetrators seemed to have found loopholes in the registration process, vulnerabilities like weak identity verification systems opened doors for fraudsters to target their victims (Gangwar et al., 2024). Consequently, it is recommended to assess its effectiveness and to propose a more secure way in the registration process to ensure safety of its users such as the use of biometrics for user verification and monitoring of SIM Cards per user (Nyarko-Boateng et al., 2022).

Several research suggested that while regulatory frameworks were implemented to protect users, it does not fully shield users from highly adaptive tactics of scammers revealing how timing and psychological manipulation played huge role in increasing victimization. International studies have already documented that SMS scams evolved through adaptive cycles. Salman et al. (2024) established that temporal variability was intrinsic to SMS scam behavior, reflecting strategic adjustment by perpetrators. Furthermore, these scam campaigns were said to exhibit cyclic behaviors, temporary sender ID (Agarwal, Suarez-Tangil, & Vasek, 2025), and identifiable lifespans before they get to adapt or shutdown (Nahapetyan et al., 2024). Through these behavioral insights, there existed an urgent need for innovation in fraud detection methods by accounting for temporal events, underscoring the temporal behavior and event-driven characteristics of SMS scams (Ghimire, 2023).

In support of this, Zimba et al. (2025) and Agarwal, Suarez-Tangil, and Vasek (2025) both agreed that scams follow patterns and routine, meticulously aligned with real-world events to reduce suspicion, as it created predictable opportunities and results. This included exploiting moments where users were expecting finances or salaries through online banking accounts, to appear legitimate and increase possibilities of victimization. Furthermore, scammers leveraged context-based opportunities, using crises or events that spark public concern to impose reliability and trust (Ramadan et al., 2021). This showed that SMS scams were planned acts that take advantage of opportunities rather than random events.

Although policies and regulations were implemented to combat SMS scam incidents, the

continuous rapid evolution of scam tactics required more data-driven and adaptive solutions. As real-world scams employed distortion of language, code-switching, and context-based contents, relying on advance models such as GPT-5 and traditional manual classifications exhibited strong multilingual robustness, making it capable of handling Filipino linguistic structures and contexts (Nery et al., 2025). Furthermore, GAMM made it capable of modeling temporal scam patterns. GAMM was able to exhibit nonlinear relationships, trends, seasonality, and random effects; rather than assuming linear relationships between variables, it was able to capture irregularities by allowing the data to assume functional form (Cabrera & Taylor, 2018). This allowed the study to include variables that potentially affect the behavior of SMS scams and offered a cornerstone and justification of using GAMM in this study.

METHODOLOGY

This chapter presents a detailed overview of the research methodology, including the study's population and sample, research instruments, data collection procedures, statistical analyses, and ethical considerations. It outlines the framework and approach used to examine the prevalence and patterns of SMS scams following the implementation of the SIM Registration Act.

Research Design

This study employed a quantitative descriptive retrospective design, supplemented by a structured observational approach, to analyze temporal patterns and behaviors in SMS scam activity. Quantitative descriptive designs allow the objective examination of trends and distributions in existing data, while retrospective designs enable the analysis of past records to identify changes over time. Structured observation further supports the systematic recording of behaviors using predefined criteria (Chauhan et al., 2024; Slater & Hasson, 2024; Thomas & Zubkov, 2023). In this study, these approaches were applied to SMS messages collected from 2023 to 2025 to examine patterns across time, including variations by hour, day, and season. This combined design enabled the study to capture both temporal dynamics and observable scam behaviors following the implementation of the SIM Card Registration Act.

Participants of the Study

The population of the study involved all college students enrolled in Bulacan State University Main Campus during the first semester of the academic year 2025–2026. From a population of 28,513 students, 50 students were selected using stratified random and purposive sampling to ensure representation across colleges while meeting the study criteria.

A sample of 50 students was chosen based on the rule of thumb in multilevel modeling, which suggested that at least 30 clusters with multiple observations per cluster were sufficient to reliably estimate fixed and random effects (Hox, 1998; Kreft & Yoon, 1994). With each participant contributing at least 400 messages, this sample size ensured adequate data to capture both individual patterns and overall trends in SMS scam behaviors.

Purposive sampling was used to select only registered Android users who do not regularly delete scam messages, as the study aimed to examine the prevalence and patterns of SMS scams following the implementation of the SIM Card Registration Act. Proportional allocation was applied based on the number of students in each college to ensure a fair distribution across colleges.

Instrument and Procedures

This study utilized an informed consent form and the SMS Backup and Restore application to extract SMS data from participants' mobile devices. The application was selected based on its data security and reliability, while event-related data (e.g., holidays and disasters) were manually compiled from relevant sources.

Prior to data collection, ethical approval was obtained from the College of Science Local Research Ethics Review Committee (ERC). Participants were screened based on eligibility criteria and were fully informed about the study through the consent process. Upon agreement, SMS data were extracted from their devices and securely transferred to a restricted-access storage system. All collected data were anonymized, and confidentiality was strictly maintained throughout the process.

Data Preprocessing

The dataset underwent preprocessing and cleaning to ensure the readiness for the analysis. Upon cleaning the dataset, the datasets were fed to DeepSeek API v3.2 Chat for classification. The classification procedure included determining whether a message was a scam or not. Scams were categorized as spoofing or smishing, while non-scam messages were categorized as promotional, transactional, or other. Each message flagged as a scam by DeepSeek was manually verified for accuracy. A criterion was set by the researchers as a basis for manual verification of scams.

After classification, the dataset was structured and normalized up to the second normal form to maintain consistency and eliminate redundancy. SMS messages served as the central table, linked to respondent information and event variables through relational keys. The finalized dataset, including temporal and event-based variables, was stored and managed using Microsoft Access for analysis.

Data Analysis

The dataset used in the study contained scam counts across different time binnings—hourly, daily, weekly, and monthly—which served as the response variable. This could take the form of prevalence (proportion) and the total count of scam messages. To account for the volume of messages received in the total count form, an exposure adjustment (offset) was introduced—ensuring that scam counts are relative and reflect scam intensity.

1. Prevalence Form

$$Prevalence_{it} = \frac{y_{it}}{n_{it}} \quad (1)$$

2. Total-count Form

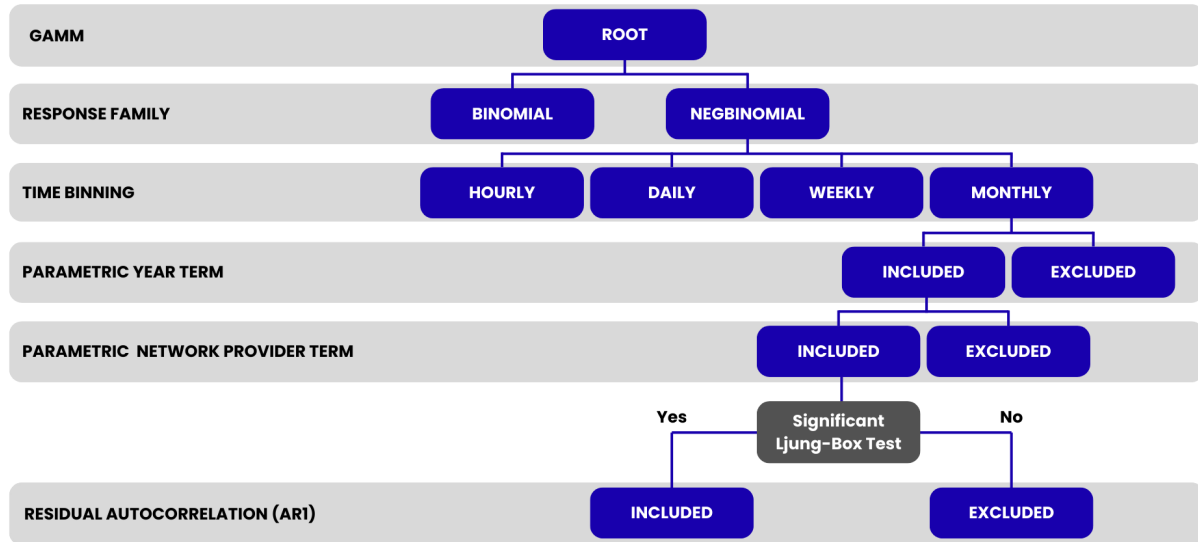
$$TotalCount_{it} = y_{it} \quad (2)$$

where y_{it} = number of scam messages received by respondent i in time bin t
and
 n_{it} = total number of messages (scam + non-scam) received by
respondent i in time bin t .

Other key explanatory variables included time, associated events, and the messages' service center route (SMSC_network). The temporal variables, including the overall trend (time_index) and cyclic terms hour-of-day (hod), day-of-week (dow), and month-of-year (moy) were derived from raw UNIX timestamps recorded in milliseconds for each SMS message and were treated as smooths. For data enrichment, the study explored events such as paydays at both the middle and end of the month (event_pay_eom, event_pay_mid), disasters (event_volcano, event_earthquake, event_typhoon_pre, event_typhoon_landfall, event_typhoon_post), and holidays (event_holiday) (Ghimire, 2023). These related events were parametric and binary in nature and took a value of 1 when a message fell within the time frame of the event. Aside from temporal variables taken as smooths, the random effect was also included to account for the inherent hierarchy and dependence in the data where multiple messages were owned by a single participant. Another form of dependence, called temporal dependence—which meant the count of scams received today could be related to yesterday's—was investigated by checking for residual autocorrelation.

Figure 1

Model Configurations



Note: The figure illustrates the 32 primary model configurations evaluated in the study. Starting from a fixed root specification, alternative configurations arise from the choice of response family, time binning level, and the inclusion or exclusion of parametric year and network provider terms. Residual autocorrelation was applied conditionally based on the Ljung–Box test.

The root was the starting point and was composed of the basic relationship between scam messages and the key explanatory variables defined. From there, the response family was chosen based on the form of the scam counts and the level of time binnings to be used.

1. Binomial–logit (prevalence)

$$Y_{ijt}^{(b)} \sim \text{Binomial} \left(p_{ijt}^{(b)} \right), \text{logit} \left(p_{ijt}^{(b)} \right) = \eta_{ijt}^{(\text{Root},b)} + \sum_{k=1}^6 \delta_k C_k \quad (3)$$

2. Negative Binomial–log (total-count)

$$Y_{ijt}^{(b)} \sim \text{NB} \left(\mu_{ijt}^{(b)}, \theta \right), \log \left(\mu_{ijt}^{(b)} \right) = \eta_{ijt}^{(\text{Root},b)} + \text{offset}(\log \text{exposure}_{ijt}^{(b)}) + \sum_{k=1}^6 \delta_k C_k \quad (4)$$

where $\eta_{ijt}^{(\text{Root},b)}$ = the root of chosen binning b and
 $\delta_k \in \{0, 1\}$ = toggles the inclusion of component C_k in each branch.

As defined in the research objectives, additional parametric terms such as the year the scam message happened (year) and the network provider of the participants (receiver_network) were tested for their association with scam counts. The combination of these choices led to the evaluation of 32 primary models under systematic sensitivity analysis.

The study utilized a method that aligned with the nonlinear behavior of the data across time, the hierarchical data structure, and residual autocorrelation, which was Generalized Additive Mixed Models (GAMM)—a natural extension of Generalized Linear Models (GLM) (Wood, 2017). To build the GAMM models, the `bam()` function in the `mgcv` package, which was suitable for R programming language versions 3.6.0 or newer, ran through RStudio version 4.5.2, was used. It allowed nonlinear relationships to be modeled while maintaining the interpretability of regression-based frameworks (Wood, 2025). Although `bam()` relied on the `gam()` fitting engine rather than the `gamm()` wrapper, it can still estimate mixed-effects structures. It does this by penalizing smooth terms using `bs = "re"`—applying a ridge-type penalty equivalent to assuming normally distributed random effects—allowing random intercepts and other grouping-level effects to be incorporated directly within the additive model.

It followed that assessing the suitability of GAMMs began by verifying whether some or all the assumptions required for GLMs were violated, thereby justifying the use of GAMM as a nonparametric extension. GLMs typically operate under the following assumptions:

1. A linear relationship between response and predictor variables is present:

$$y_i = \beta_0 + \beta_1 \times x_i + \epsilon_i \quad (5)$$

2. The error is normally distributed:

$$\epsilon_i \sim N(0, \sigma^2) \quad (6)$$

3. The variance of the error is homogeneous (homoscedastic)
4. The errors are independent of each other.

While nonlinearity was expected and modeled through smooth terms, GAMMs still retained key assumptions regarding the use of appropriate family distribution and corresponding link function. However, normality, homoscedasticity, and independence of errors were still expected. Exploratory

data analysis provided necessary descriptive statistics but also uncovered the following: (1) nonlinearity through scatterplots and LOESS smooths, (2) overdispersion through the ratio of the variance and mean, and (3) autocorrelation through Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots.

The model space was constructed by combining response family, time binning, year inclusion, and receiver network inclusion, yielding 32 primary specifications. Residual autocorrelation was then assessed using the Ljung–Box test. A significant result indicated the need for an AR(1) adjustment (Noble, n.d.), which, when applied, conditionally extended the evaluation to 64 fitted models while preserving the original 32-model design. The models were further evaluated through the DHARMA package, which used simulation-based residuals to assess departures from model assumptions (Hartig, 2024). The uniformity property served as the central diagnostic criterion in the evaluation process. It tested if the simulated residuals significantly deviate from the expected uniform distribution. A non-significant result ($p > 0.05$) indicated no evidence against the assumed distribution, while a significant result ($p < 0.05$) suggested possible distributional misspecification. Models that failed this primary distributional check were not considered suitable for interpretation. After establishing distributional adequacy through the uniformity test, additional residual diagnostics were assessed. These include dispersion, outliers, and zero inflation. For each of these tests, the null hypothesis stated that the model was correctly specified, while significant results indicated potential misspecification, prompting further evaluation and caution in interpretation. Furthermore, other metrics were measured: (1) residual autocorrelation through Ljung-Box test, (2) smooth adequacy through the effective degrees of freedom (EDF) and k-index, and (3) concurvity for redundancy among smooth terms. Through this structured diagnostic process, conclusions were based on nuanced, interpretable, statistically valid, and well-specified models.

GAMM estimates were transformed into percentage changes for interpretability. Given the log link under the negative binomial family, effects were interpreted on the log scale and converted as:

$$\%change = (e^{\beta} - 1) \times 100 \quad (7)$$

for parametric event effects, representing the change in expected scam counts relative to the baseline.

For smooth terms, percentage changes between selected time points were computed using:

$$\%change = (e^{\eta_b - \eta_a} - 1) \times 100 \quad (8)$$

where η_a and η_b denote the estimated values of the smooth at two time points. This reflected the multiplicative change in expected scam counts between the selected points. All reported percentages were derived from these transformations.

Ethical Consideration

This study strictly followed ethical standards to protect the rights, privacy, and well-being of all participants, who were all college students enrolled at BuSU Main Campus during the first semester of academic year 2025-2026. Before data collection, informed consent was obtained, and participants were fully informed about the study's purpose, procedures, their rights, and the confidential handling of

their data. Participation was voluntary, and respondents could withdraw at any time without penalty.

Since the study involved extracting SMS messages, potential privacy risks were addressed by allowing participants to review and delete any messages they did not wish to share. The application only accessed SMS data and did not retrieve other personal files, images, or accounts. All collected data was anonymized, coded, and stored securely, with identifying information removed. Data was accessible only to the researchers and would be permanently deleted after three years or upon request.

In accordance with RA 10173 (Data Privacy Act of 2012), consent from the device owner was considered sufficient, as third-party sender information was anonymized and not treated as human subject data. As all participants were legally independent adults, no additional consent was required. The research protocol was reviewed and approved by the BulSU College of Science Ethics Review Committee, ensuring compliance with institutional and national data privacy standards.

RESULTS AND DISCUSSIONS

This chapter presents the results of the study and their corresponding interpretations based on the data gathered from the students. The study aims to determine the prevalence and patterns of SMS scams following the implementation of the SIM Card Registration Act among college students at BulSU. The collected data were organized and analyzed using appropriate statistical tools to describe the patterns and relationships among the variables. Specifically, the chapter discusses the origins and distribution of SMS scams, temporal scam patterns, the influence of specific events on scam activity, the robustness of the models used, and the implications of the findings for SMS scam mitigation.

Descriptive Analysis of SMS Scams among BulSU Students

A total of 102,311 messages were collected from 50 college students between July 25, 2023 and December 31, 2025. Out of these numbers, 3,267 messages (3.20%) were classified as SMS scams. As shown in Table 1, scam messages represented a small proportion of the dataset relative to promotional, transactional, and other messages. However, their absolute count remains substantial, indicating that fraudulent messaging continues to occur despite the implementation of the SIM Card Registration Act. SMS scams persisted despite the implementation of the SIM Card Registration Act, indicating that the policy has not fully eliminated fraudulent messaging. Similar policies were shown to fall short due to exploitable weaknesses, which highlighted the need to reassess their effectiveness (Gangwar et al., 2024; Nyarko-Boateng et al., 2022).

Table 1*Classification of Messages*

Message Type	n	Percentage
Promotional	33861	33.1%
Others	32590	31.9%
Transactional	32575	31.8%
Scam	3267	3.2%
Total	102331	100%

Among the identified scam messages, 57.1% were classified as smishing, while 42.9% were spoofing. As shown in Table 2, smishing accounts for the majority of scam activity. The dominance of smishing highlights a policy gap, as the current law primarily targets spoofing, allowing content-based deception to persist.

Table 2*Attack Type Distribution*

Attack Type	n	Percentage
Smishing	1867	57.1%
Spoofing	1400	42.9%

In terms of origin, SMART accounted for the largest volume of scam messages, while “Other” SMSC networks exhibited the highest scam rate at 15.1%. As shown in Table 3, the “Other” SMSC networks category suggested that scam messages were frequently routed through unidentified or irregular transmission channels. The pattern pointed to the use of compromised or unauthorized devices, consistent with reports of illegal devices intercepting and mimicking cell towers to distribute fraudulent messages (Guiao, 2024).

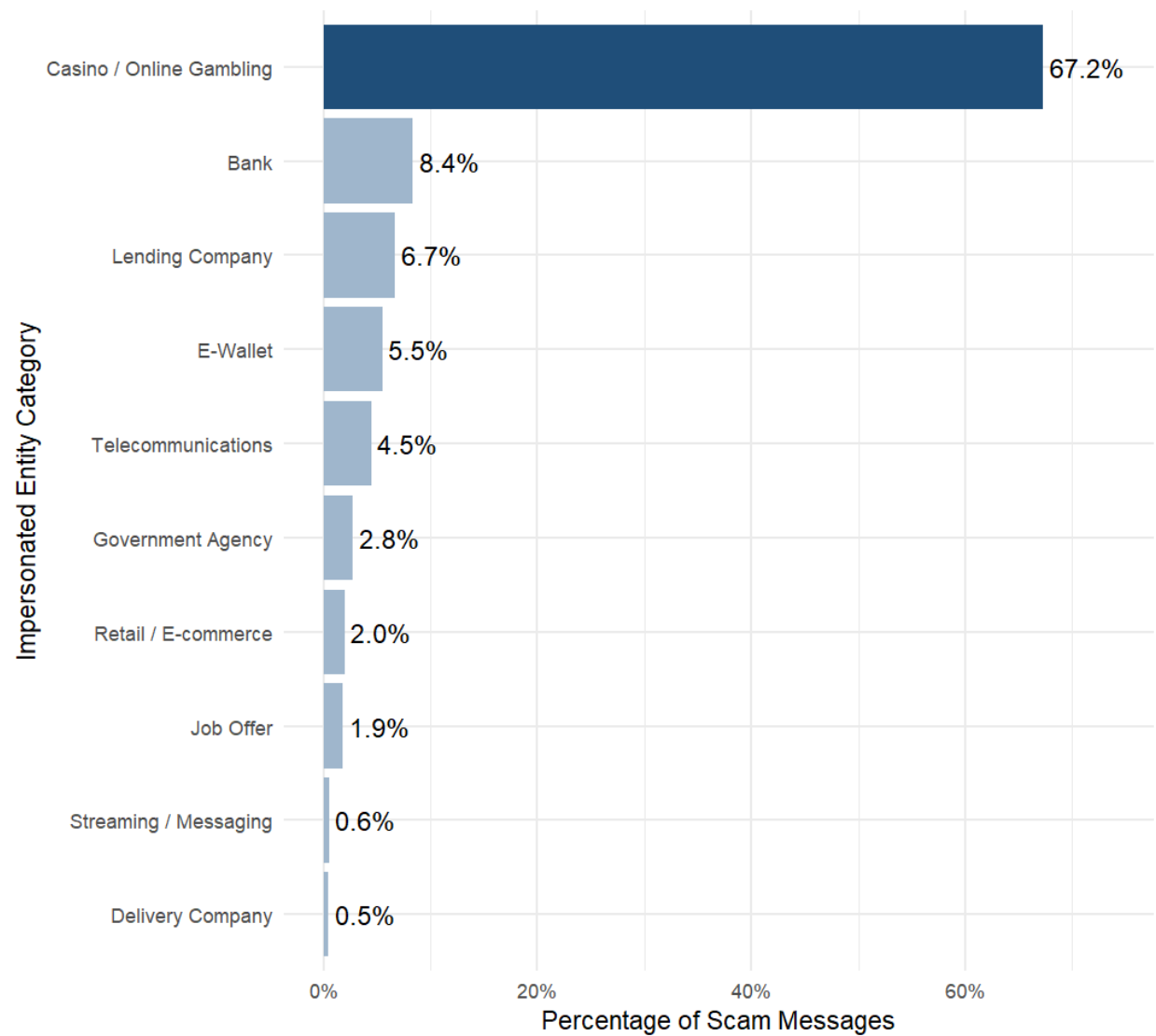
Table 3*Origin of SMS Scams According to Telecommunication Network*

SMSC Network	Scam Message	Total Message	Proportion
Other	135	892	15.1%
SMART	2859	66230	4.3%
GLOBE	243	27348	0.9%
DITO	24	5029	0.5%
Null	6	2812	0.2%

Spoofed messages were most associated with casino and online gambling platforms,

accounting for approximately 67% of impersonation cases, followed by banks and lending services. As shown in Figure 2, this pattern reflected finance-related targeting, where scammers impersonate entities associated with money to increase message credibility and engagement. This aligned with findings that scam messages frequently imitate financial institutions and continuously adapt their content to appear legitimate (Agarwal, Suarez-Tangil, Vasek, et al., 2025; Dela Cruz, 2025; Zimba et al., 2025).

Figure 2
Impersonated Entities of Spoofed Messages



Across receiver networks, DITO users experienced the highest relative exposure to scam messages at 5.4%, followed by SMART and GLOBE. As shown in Table 4, this variation indicated that exposure differs across networks despite all providers being affected. The pattern reflected how scam strategies adapt to system vulnerabilities rather than remaining fixed, consistent with findings

that scam methods continuously evolve over time (Abraham et al., 2024).

Table 4

Distribution of Scams by Receiver Networks

Receiver Network	Total Messages	Scam Messages	Prevalence
DITO	9600	520	5.4%
SMART	63803	2396	3.8%
GLOBE	28908	315	1.2%

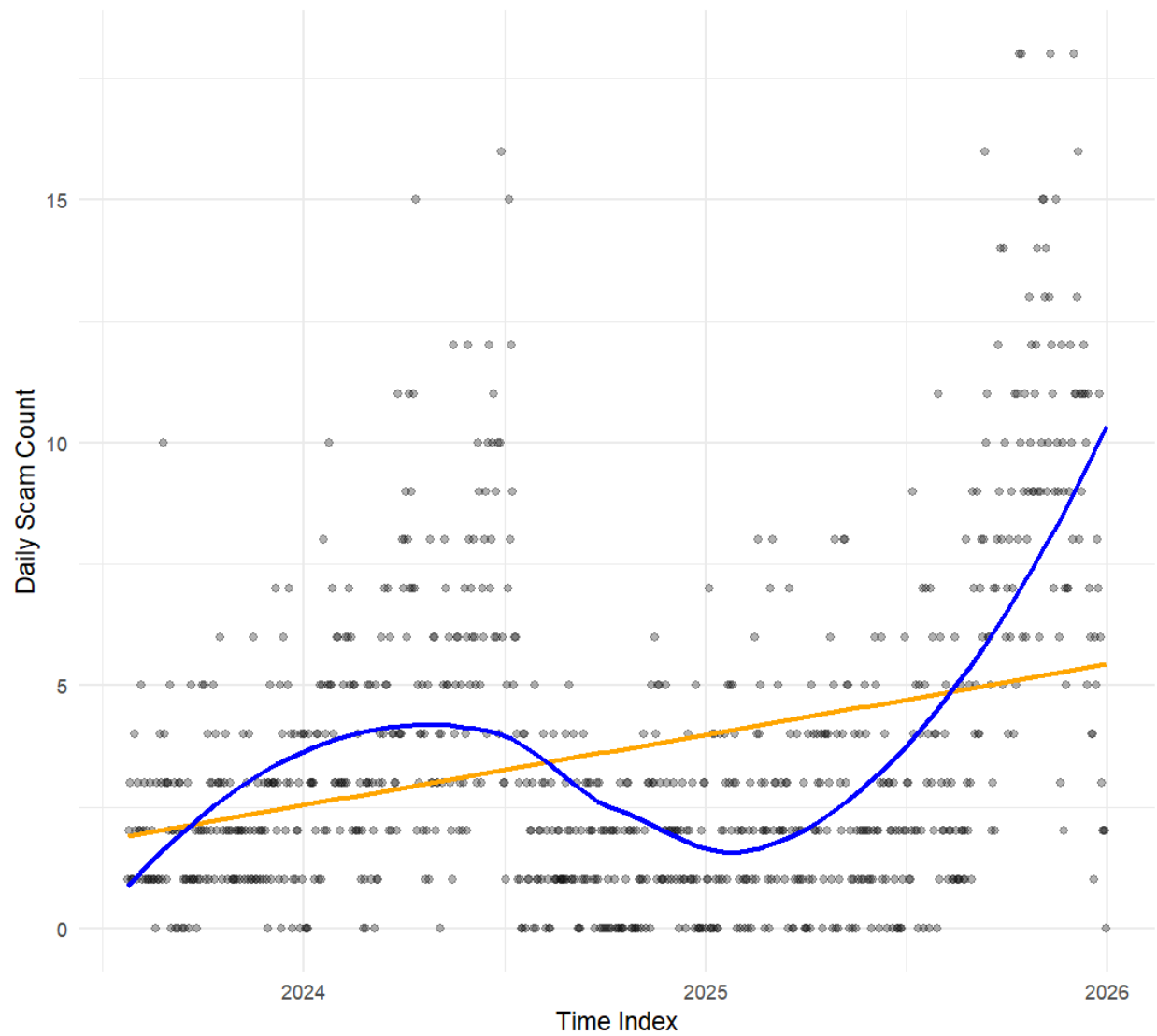
Overall, the descriptive results showed that SMS scams remain present, were dominated by smishing strategies, frequently utilized irregular transmission channels, and concentrated on financially themed impersonation. These patterns indicated adaptive behavior that exploited both policy gaps and system-level vulnerabilities, providing a foundation for the subsequent temporal and model-based analyses.

Assessing GAMM Validity

The study employed Generalized Additive Mixed Models (GAMM) to analyze temporal patterns in SMS scam activity, and model validity was first established before any interpretation. As shown in Figure 3, the data exhibited clear nonlinear structure and substantial dispersion, supporting the use of GAMM and justifying the selection of the negative binomial family while excluding the Poisson distribution due to its restrictive variance assumption. All 32 model configurations converged successfully.

Figure 3

Overall Temporal Trend in Daily Scam Counts



However, Table 5 showed that models without AR(1) consistently failed the Box–Ljung test ($p < .001$), indicating unresolved temporal dependence. Once AR(1) was incorporated, the same table demonstrated substantial improvements in model fit, reflected in lower AIC and BIC values and higher log-likelihood estimates across all specifications.

Table 5*Summary of Model Fit Improvements After AR(1) Inclusion*

Family	Binning	Δ logLik	Δ AIC	Δ BIC	Remarks
Negative Binomial	Hourly	49.99%	49.73%	48.31%	Marginal Penalization
	Daily	12.59%	12.50%	12.15%	Loss and Consistent
	Weekly	11.56%	11.46%	11.15%	Scaling
	Monthly	5.40%	5.37%	5.29%	
Binomial	Hourly	9.40%	9.33%	9.05%	Marginal Penalization
	Daily	10.42%	10.35%	10.09%	Loss and Inconsistent
	Weekly	10.87%	10.74%	10.34%	Scaling
	Monthly	1.81%	1.73%	1.54%	

Further validation using Table 6 indicated that only the negative binomial models at the daily, weekly, and monthly levels satisfied the primary uniformity condition, while binomial models consistently failed and were therefore excluded from interpretation. The same table showed that although overdispersion persisted at finer resolutions, particularly in daily models, all valid specifications passed the zero-inflation test, indicating that the observed variability reflected inherent fluctuations in scam activity rather than structural misspecification. Overall, these results established that negative binomial models at the daily, weekly, and monthly levels provided a statistically valid basis for interpretation, while hourly results were retained only for cautious comparison.

Table 6*Summary of Model Validity through Residual Diagnostics*

Family	Binning	Uniformity	Dispersion	Outlier	Zero Inflation	Remarks
Negative Binomial	Hourly	0%	0%	0%	0%	Invalid
	Daily	100%	0%	50%	100%	Valid
	Weekly	100%	88%	88%	100%	Valid
	Monthly	100%	100%	100%	100%	Valid
Binomial	Hourly	0%	50%	0%	50%	Valid
	Daily	0%	0%	0%	50%	Invalid
	Weekly	0%	0%	0%	0%	Invalid
	Monthly	0%	0%	0%	0%	Invalid

Beyond establishing validity, systematic sensitivity analysis evaluated whether these results remained stable across alternative specifications. Table 5 demonstrated that the inclusion of AR(1), receiver network effects, and other structural components consistently improved model fit across temporal binnings, with gains in log-likelihood accompanied by corresponding improvements in

penalized criteria such as BIC. This alignment indicated that added components contributed meaningful explanatory power rather than unnecessary complexity. Across temporal resolutions, the same table showed that negative binomial models exhibited consistent scaling in improvements, while binomial models displayed irregular patterns, reinforcing their misalignment with the data structure.

At the same time, Table 6 showed that model stability improved with temporal aggregation, as dispersion and outlier issues decreased from daily to weekly and monthly levels. Together, these results confirmed that the identified temporal patterns were robust to reasonable modeling choices, supporting the use of the daily negative binomial specification with AR(1) as the primary reference model for interpretation.

Temporal Patterns and GAMM-Based Analysis of SMS Scam Activity

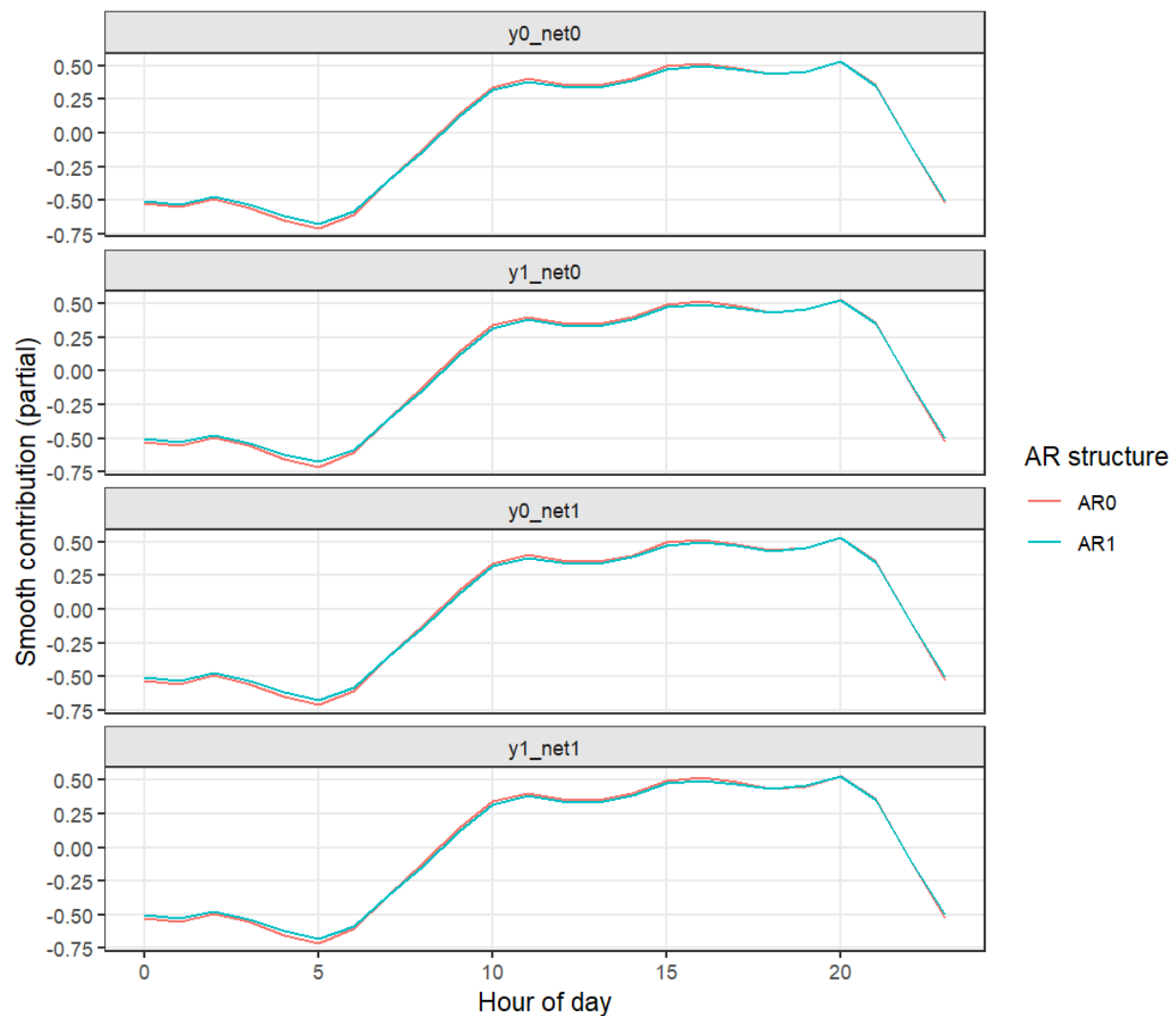
Descriptive time plots provided an initial view of temporal patterns in SMS scam activity. In this section, these patterns are explicitly evaluated against GAMM results to determine whether they remain statistically significant when formally modeled. Variables that appeared relevant in the descriptive time plots did not always remain important once incorporated into the model. The receiver network, while it showed differences in scam exposure descriptively, did not exhibit statistically significant baseline effects in the GAMM. In contrast, the respondent-level random intercept was consistently significant, indicating that variation in scam exposure is better explained by individual-level differences rather than network-level differences. Similarly, the linear year term was generally not significant, and when it appeared so, the effect was associated with models that did not account for residual autocorrelation. This suggested that apparent year effects reflected noise disguised as real patterns.

To address this, interpretation was anchored on a single reference specification selected from among the diagnostically acceptable models. Multiple configurations were estimated across alternative family distributions, binning levels, and structural components, but inference required a stable and interpretable model. The daily negative binomial model was chosen as the primary reference, as negative binomial appropriately modeled count data while daily aggregation preserved meaningful temporal variation without introducing excessive short-term noise. This specification satisfied the required residual diagnostics and included key structural components such as receiver network effects and AR(1) correction. Alternative specifications were used to assess robustness, but interpretation proceeded from this reference model to ensure consistency and statistical validity.

Hourly patterns indicated that scam activity was minimal during early morning hours, reaching its lowest point at around 5 AM, before rising sharply beginning at 8 AM and peaking at approximately 11 AM, with sustained activity throughout the afternoon and another increase during the evening period. A corresponding pattern was reflected in Figure 4, where the estimated smooth rose beginning at around 5 AM and reached a peak at approximately 10 AM, corresponding to an increase of about 170% relative to early morning levels. The close agreement between the descriptive pattern and the model-based estimate indicated that scam activity was systematically concentrated during periods of high user engagement rather than occurring randomly through the day, consistent with patterns of opportunistic targeting where attackers exploited predictable routines in user behavior (Agarwal, Suarez-Tangil, & Vasek, 2025).

Figure 4

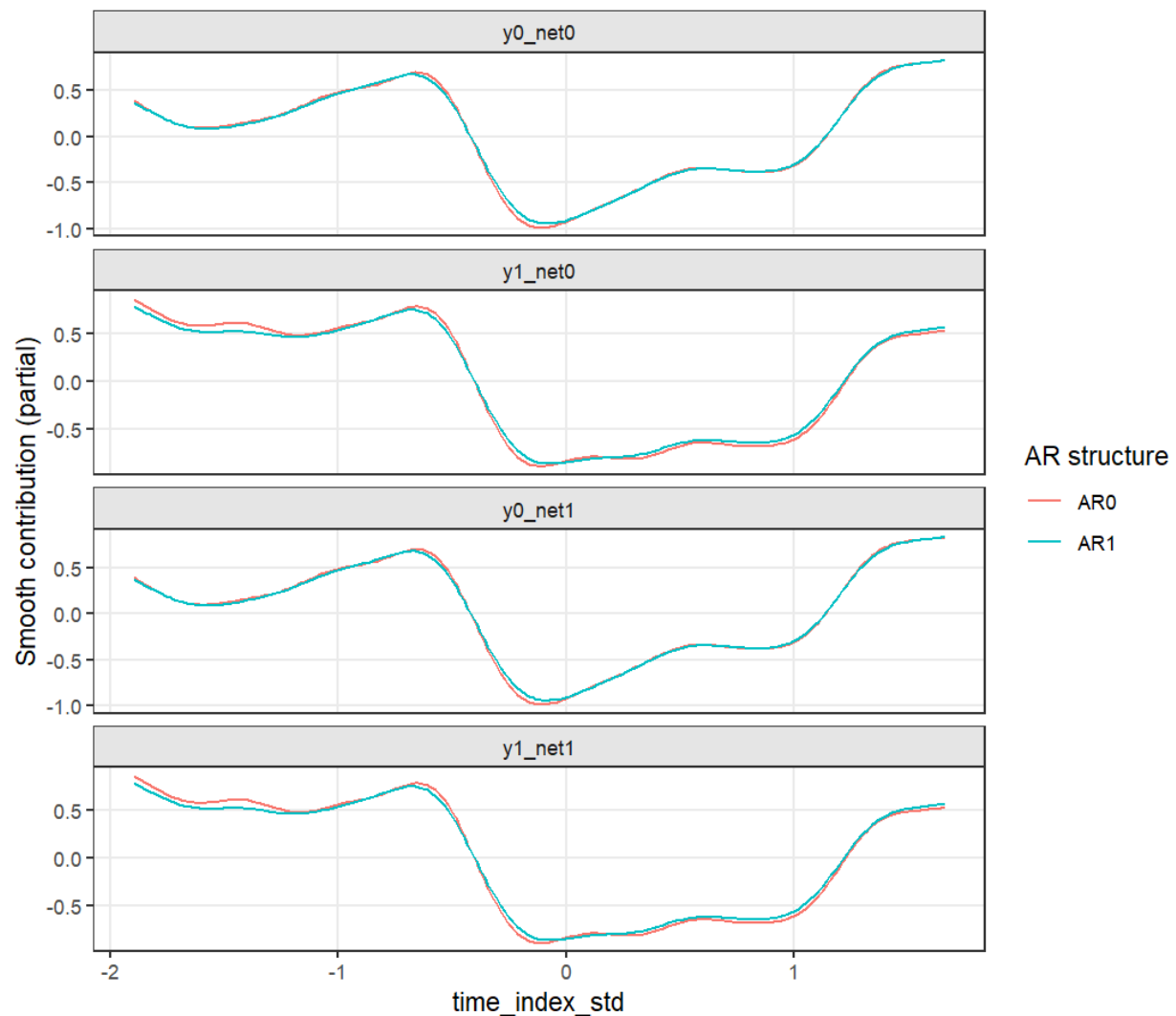
Hourly Negative Binomial Hour-of-Day Smooth by Model Configuration



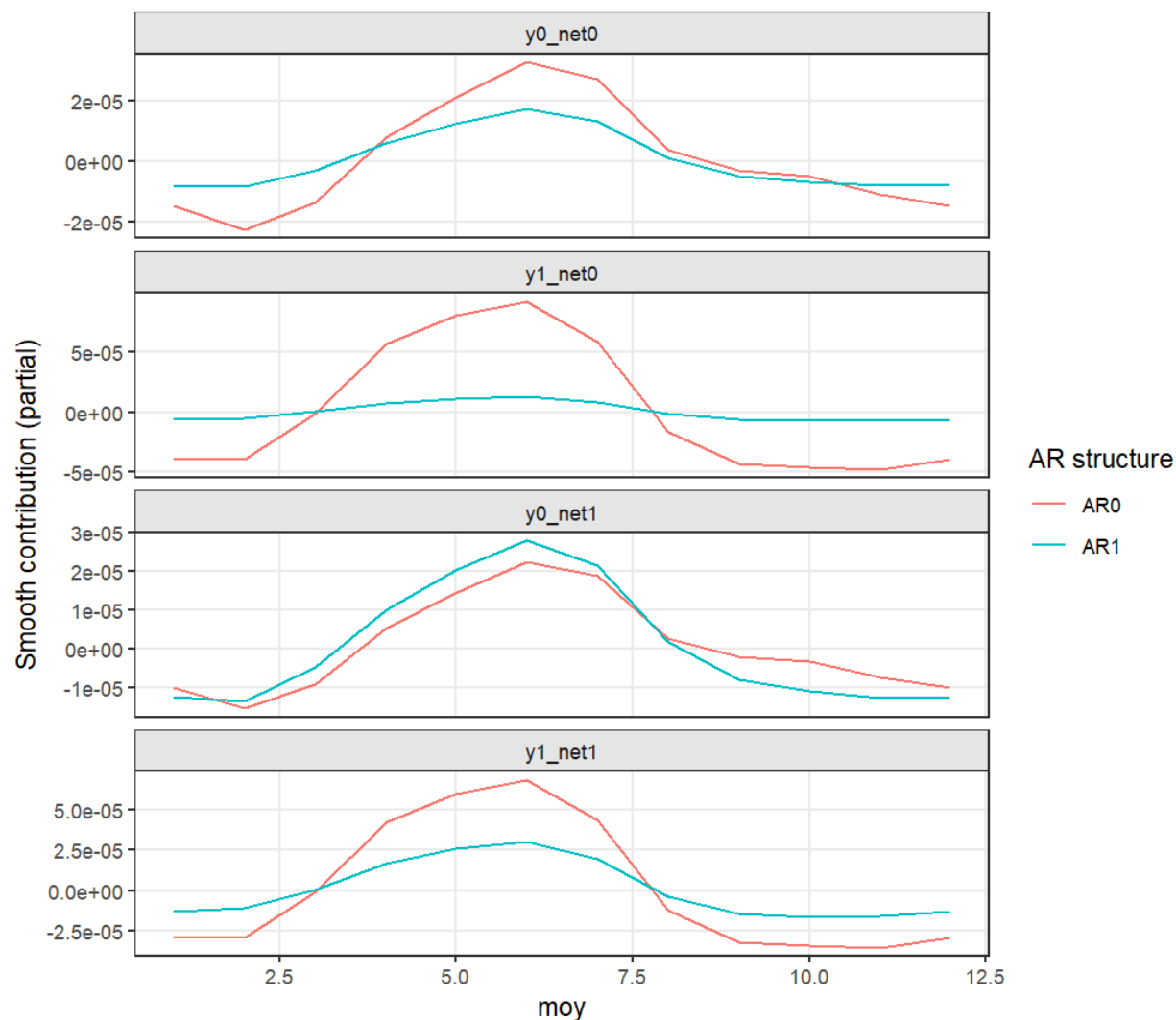
Daily patterns suggested higher average scam prevalence on Mondays and Saturdays, with relatively lower activity on Fridays and Sundays, as reflected in Figure 5. However, the corresponding GAMM smooth for day of the week was consistently penalized toward zero, indicating that these differences were not statistically stable once temporal dependence was accounted for. This showed that scam activity does not meaningfully differ across weekdays, despite the apparent variation in descriptive plots. The lack of a stable weekly structure suggested that scammers operated continuously rather than following fixed schedules, aligning with theoretical perspectives that emphasized flexible exploitation of opportunities rather than rigid temporal routines (Brantingham & Brantingham, 2021; Cohen & Felson, 1979).

Figure 5

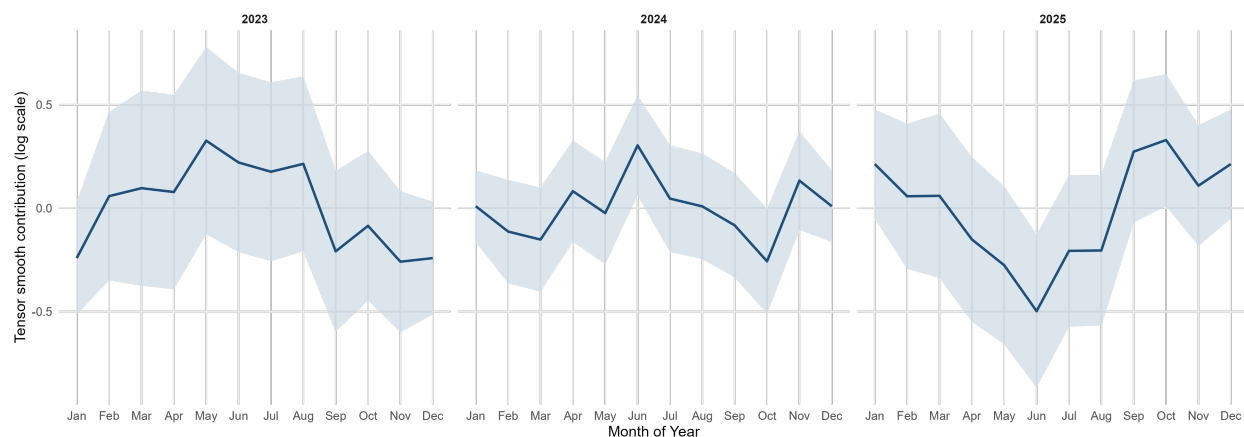
Daily Negative Binomial Trend Smooth by Model Configuration



Monthly patterns showed a clear peak in June, with elevated activity from March to June, followed by a decline in July and August and another rise during the pre-holiday period, as seen in Figure 6. These peaks suggested that scam activity increases during specific parts of the year. However, the corresponding GAMM smooth for month of year does not retain this pattern consistently, as it was unstable across specifications and often shrunk toward zero. This indicated that while peaks were visible in the data, they do not occur at the same time every year, suggesting that the pattern was not a fixed seasonal cycle.

Figure 6*Daily Negative Binomial Month-of-Year Smooths by Model Configuration*

To examine this further, a tensor-based smooth was introduced to model the interaction between month of year and the overall time trend. This was not only based on the instability of the month-of-year smooth, but also on the strong overlap between seasonal and long-term effects. Earlier results showed that month of year and time_index share a large amount of variation, making it difficult for a single smooth term to separate fixed seasonality from changes over time. The tensor allowed these components to vary together instead of forcing them into one structure. The resulting pattern, shown in Figure 7, made the shift clearer: activity peaked around June in 2024, but in 2025 the peak moved later, around October. This showed that the observed peaks were not fixed seasonal effects but changed over time. Consistent with this, Table 7 showed only small changes in overall fit, indicating that the tensor mainly clarified structure rather than improving performance. This supported the interpretation that scam activity followed shifting waves instead of repeating at the same time each year, reflecting adaptive timing rather than fixed seasonality (Nahapetyan et al., 2024; Salman et al., 2024).

Figure 7*Daily Negative Binomial Month-of-Year by Year Tensor Smooth***Table 7***Model Comparison for Reference Model With and Without Tensor Smooth*

Metric	AR1 (Time Smooth Only)	AR1 (Time + Tensor)
logLik	-6840.107	-6853.003
AIC	13817.95	13865.21
BIC	14409.95	14549.47
Deviance Explained (%)	36.1	36.5
Adjusted R^2	0.1031	0.1070
Dispersion (DHARMA p)	0.000 (FAIL)	0.020 (FAIL)
Uniformity (DHARMA p)	0.984 (OK)	0.5141 (OK)
Outlier Test (bootstrap p)	0.000 (FAIL)	0.000 (FAIL)
Zero Inflation (p)	0.120 (OK)	0.150 (OK)
Ljung-Box (std.rsd p)	AC (0.000)	AC (0.000)

The long-term trajectory provided a broader temporal context for these patterns. The trend shown in Figure 7 rose toward the second quarter of 2024, followed by a sharp decline of approximately 80–83% into early 2025, and then a substantial rebound toward late 2025. This decline aligned with intensified government action and the subsequent ban on POGO operations, indicating a disruption in scam-related activity during this period. However, the rebound showed that this reduction was temporary, as scam activity resurged after operators adjusted their strategies. The pattern demonstrated that scam activity responded to regulatory interventions but adapted over time, resulting in temporary suppression rather than sustained reduction, consistent with evidence that scam campaigns operated in adaptive cycles under enforcement pressure (Agarwal, Suarez-Tangil, & Vasek, 2025).

Events Affecting SMS Scams

Event effects were most consistent within the negative binomial family under daily binning, as shown in Table 8. Across model specifications, salary release periods, holidays, volcanic activity, and post-typhoon recovery were significantly and positively associated with increased scam counts. Even events that were not statistically significant, such as earthquakes, pre-typhoon periods, and landfall, showed consistently positive coefficients, although their contribution to explained variance was small.

This consistency across specifications indicates that the relationship between events and scam activity was not driven by specific modeling assumptions. Whether autocorrelation was accounted for through AR(1), or whether receiver network and year terms were included, the direction and significance of key event variables remained stable. At finer temporal resolution, particularly in hourly models, widespread significance was observed, likely reflecting higher variability rather than stronger substantive effects. As the data were aggregated into weekly and monthly binnings, event effects weakened and statistical significance decreased, which was expected since event indicators were defined at the daily level and become diluted under aggregation.

In the reference daily negative binomial model, the estimated dispersion parameter ($\theta = 0.847$) indicated that scam activity occurred in clustered bursts rather than at a constant rate. The intercept (-4.64308) represented the baseline expected scam count when no event was present, and smooth terms were centered, corresponding to roughly one scam message every 100–110 days per respondent. This established a low baseline level of activity outside event periods.

Table 9

Estimated Event Effects from the Reference Daily Negative Binomial Model

Term	Estimate	Equivalent Percentage	Std. Error	t	p
Intercept	-4.643	-	0.723	-6.43	< 0.001*
event_pay_mid	0.329	39%	0.064	5.16	< 0.001*
event_pay_eom	0.258	29%	0.064	4.03	< 0.001*
event_holiday	0.274	32%	0.089	3.07	0.002*
event_earthquake	0.149	16%	0.246	0.61	0.544
event_volcano	0.408	50%	0.153	2.67	0.007*
event_typhoon_pre	0.081	8%	0.104	0.78	0.437
event_typhoon_landfall	0.201	22%	0.114	1.76	0.078
event_typhoon_post	0.407	50%	0.086	4.72	< 0.001*

Note: (*) Indicates that the event is significant.

Relative to this baseline, mid-month salary release was associated with an increase of approximately 39% in expected scam messages, while end-of-month salary release corresponded to an increase of about 29%. Holiday periods were associated with an increase of approximately 32%. The largest effects occurred during volcanic activity and post-typhoon recovery, each corresponding to roughly a 50% increase in expected scam messages. Although earthquakes, pre-typhoon periods,

Table 8*Direction and Significance of Event Effects Across Model Specifications*

Model	Pay_mid	Pay_eom	Holiday	Earthquake	Volcano	Typhoon_pre	Typhoon_landfall	Typhoon_post
y0_net0_ar0	↑ 0.000 *	↑ 0.000 *	↑ 0.000 *	↑ 0.140	↑ 0.002 *	↑ 0.460	↑ 0.055	↑ 0.000 *
y0_net0_ar1	↑ 0.000 *	↑ 0.000 *	↑ 0.002 *	↑ 0.544	↑ 0.007 *	↑ 0.437	↑ 0.078	↑ 0.000 *
y0_net1_ar0	↑ 0.000 *	↑ 0.000 *	↑ 0.000 *	↑ 0.140	↑ 0.002 *	↑ 0.460	↑ 0.055	↑ 0.000 *
y0_net1_ar1	↑ 0.000 *	↑ 0.000 *	↑ 0.002 *	↑ 0.544	↑ 0.007 *	↑ 0.437	↑ 0.078	↑ 0.000 *
y1_net0_ar0	↑ 0.000 *	↑ 0.000 *	↑ 0.000 *	↑ 0.156	↑ 0.002 *	↑ 0.438	↑ 0.052	↑ 0.000 *
y1_net0_ar1	↑ 0.000 *	↑ 0.000 *	↑ 0.002 *	↑ 0.553	↑ 0.007 *	↑ 0.404	↑ 0.072	↑ 0.000 *
y1_net1_ar0	↑ 0.000 *	↑ 0.000 *	↑ 0.000 *	↑ 0.156	↑ 0.002 *	↑ 0.438	↑ 0.052	↑ 0.000 *
y1_net1_ar1	↑ 0.000 *	↑ 0.000 *	↑ 0.002 *	↑ 0.553	↑ 0.007 *	↑ 0.405	↑ 0.072	↑ 0.000 *

Note: The arrow corresponds to the direction (positive or negative) of the estimates of metrics.

(*) Indicates that the estimate is significant.

Model naming convention are constructed as follows, y = year, net = provider, ar = autocorrelation; 0 = excluded, 1 = included.

and landfall were not statistically significant, their coefficients remained positive across daily models, indicating weaker but consistent directional effects.

While prior studies have documented that SMS scams respond to contextual triggers, localized empirical evidence linking specific real-world events to scam intensity in the Philippine setting remained limited. The observed increases during salary release periods were consistent with Zimba et al. (2025), who identified payday cycles as predictable windows of financial vulnerability. Holiday-related increases similarly aligned with the opportunistic patterns described by Salman et al. (2024), while disaster- and recovery-related effects reflected the crisis exploitation mechanisms discussed by Ramadan et al. (2021). These results showed that scam activity increased during financially sensitive and disruptive periods. The consistency of effects across models and the magnitude of increases relative to baseline indicated that these were not random fluctuations, but stable patterns driven by changes in exposure and opportunity.

Implications of the Findings of the Study

With the established results of the study, several insights can be made with regards to the enhancement of the policy, awareness campaigns, and strengthening network security. The following are the implications of the study:

1. **Enhancements in SIM Card Registration Policy.** The findings indicated that the current SIM registration measures remained insufficient, particularly due to the higher prevalence of smishing relative to spoofing, suggesting regulatory gaps allowed misuse of multiple SIM cards and unregulated messaging channels. This aligned with the evidence that policy limitations and weak enforcement reduced effectiveness, emphasizing the need to expand coverage to smishing, impose stricter SIM ownership controls, and simplify reporting mechanisms to improve user participation and enforcement efficiency (Abraham et al., 2024; Gangwar et al., 2024).
2. **Public Awareness Campaigns.** Observed temporal patterns indicated that scam activity was concentrated during specific high-risk periods (e.g., morning peaks, start of the week, and mid-year months), suggesting that awareness strategies should shift from general campaigns to time-targeted interventions. Public advisories and educational initiatives may be more effective when released during identified peak periods and when aligned with common scam tactics, as these can help enhance user vigilance and contextual awareness regarding when and how scams were most likely to occur (Agarwal, Suarez-Tangil, & Vasek, 2025).
3. **Network-level Security Measures.** The persistence and spread of scams across different networks, including the higher exposure observed in certain providers and the use of unauthorized transmission channels, emphasized the need for coordinated and adaptive network-level defenses. This indicated the importance of strengthening real-time detection, improving filtering systems, and promoting intelligence sharing across networks to address evolving scam strategies. These measures were necessary, as existing policies alone may not be sufficient to fully prevent fraud, especially when scammers continuously adapt their methods to bypass current safeguards (Nyarko-Boateng et al., 2022; Tumang, 2025).

Limitations of the Study

This study was limited by its use of a student-only sample, which might not represent other groups such as older users or people from different regions. In addition, some data may be incomplete because participants were allowed to delete personal messages before data collection. This reduced the total message volume in some cases, which might affect the computed prevalence of scam messages. However, the large size of the dataset helped reduce the impact of this limitation. Furthermore, the study was constrained by computational limitations, particularly in the execution of simulation-based residual diagnostics. Due to resource constraints, the number of simulations for the hourly models was reduced, which may limit the precision of diagnostic results.

Future Research Directions

Future research may expand the scope of the study by including non-Android users and a wider range of age groups to improve generalizability. Beyond temporal patterns, further studies may also examine spatial patterns of SMS scams and explore how scam message content aligns with specific events. Methodological extensions may include the use of larger datasets, longer study periods, and alternative modeling approaches, including detection frameworks that incorporate temporal features. Prior work suggests that time-linked variables are informative for fraud analysis (e.g., Ghimire (2023)), supporting the integration of temporal factors into detection models. Future researchers are encouraged to utilize or apply for access to high-performance computing systems to support more complex analyses as these extensions may require high computational resources. Finally, future studies may investigate the broader effects of SMS scams on students and assess whether scam activity persists following new regulatory measures.

CONCLUSION AND RECOMMENDATIONS

This study contributes to the fields of cybersecurity and policy evaluation by providing empirical evidence on the prevalence and temporal patterns of SMS scams among BulSU students after the implementation of the SIM Card Registration Act. By applying Generalized Additive Mixed Models (GAMM), the study shows how scam activity evolves over time and responds to real-world events, offering an interpretable framework for assessing the effectiveness of cybersecurity policies within the studied population. These findings also support the United Nations Sustainable Development Goal (SDG) 16 by contributing to stronger institutions, improved digital security, and more effective strategies against fraud.

Conclusion

This section presents the conclusions of the study based on the major findings and analysis of the data. The conclusion summarizes the key results about the prevalence and patterns of SMS scams after the implementation of the SIM Card Registration Act among college students in BulSU. These conclusions are drawn from the evidence gathered in the study and are intended to address the research objectives, as presented below:

1. The most frequent origin of SMS scams was linked to hacked cell towers or “Other SMSC networks.” Scammers mostly impersonated casino/online gambling platforms, followed by banks and lending companies.
2. In terms of exposure, DITO users were more likely to receive SMS scams compared to other Telco networks.
3. GAMM results showed that SMS scam activity follows structured temporal patterns rather than occurring randomly. The trend reflects periods of increase, decline, and rebound, indicating that scam intensity changes over time.
4. Event-based analysis showed that increases in scam counts were associated with salary release periods, holidays, and disaster recovery phases. These effects remained consistent across model specifications.
5. Overall, SMS scams operated as organized and adaptive campaigns that responded to real-world conditions. Effective mitigation therefore required timely and adaptive strategies, improvements in addressing smishing within the SIM Card Registration Act, and stronger network-level coordination and detection systems.

Recommendations

Based on the findings of the study, the following recommendations were proposed to help reduce the prevalence of SMS scams and improve the effectiveness of current policies and network practices in the Philippines. These suggestions were directed toward government agencies, telecommunication providers, and other relevant stakeholders involved in cybersecurity and public protection.

1. The SIM Card Registration Act may be expanded to address smishing, including stricter SIM verification, improved monitoring of suspicious SMS activity, and appropriate penalties for misuse.
2. Authorities are recommended to further strengthen the regulation and monitoring of SIM card resellers to prevent illegal acquisition and misuse of SIM cards.
3. A limit on the number of SIM cards that an individual can register may be implemented to reduce large-scale scam operations.
4. Telecommunication networks are recommended to continuously improve their security systems to prevent unauthorized access, including the use of illegal or hacked cell towers.
5. Telco networks might consider applying adaptive security measures and increase user awareness during peak scam periods (e.g., specific hours, days, and months) to reduce potential victimization.

REFERENCES

Abraham, J., Rogers, S., Njoki, C., & Greening, J. (2024). *[gasa] state of scams in the philippines - 2024* (tech. rep.) (Slide 10. Majority of scams are delivered via text/SMS messages & social media.Slide 7. Frequency.). Global Anti-Scam Alliance.

- Agarwal, S., Suarez-Tangil, G., & Vasek, M. (2025). An overview of 7726 user reports: Uncovering sms scams and scammer strategies [Explored time as a variable]. <https://doi.org/10.14722/ndss.2026.240987>
- Agarwal, S., Suarez-Tangil, G., Vasek, M., Harvey, E., & Mariconti, E. (2025). *'hey mum, i dropped my phone down the toilet': Investigating hi mum and dad sms scams in the united kingdom i dropped my phone down the toilet': Investigating hi mum and dad sms scams in the united kingdom*. <https://www.usenix.org/conference/usenixsecurity25/presentation/agarwal-sharad> Heymum,
- Albert, J. R., Quimba, F. M., Tabuga, A., Mirandilla-Santos, M. G., Rosellon, M. A., Vizmanos, J. F., Cabaero, C., & Muñoz, M. (2021). Expanded data analysis and policy research for national ict household survey 2019.
- Brantingham, P., & Brantingham, P. (2021, January). Crime pattern theory. <https://doi.org/10.1093/acrefore/9780190264079.013.8>
- Cabrera, M., & Taylor, G. (2018). Modelling spatio-temporal data of dengue fever using generalized additive mixed models. *Spatial and Spatio-temporal Epidemiology*, 28, 1–13. <https://doi.org/10.1016/j.sste.2018.11.006>
- Chauhan, L., Bunkar, R., Verma, A., & Mounika, B. (2024, November). Observational research: A qualitative research method.
- Cohen, L. E., & Felson, M. (1979, August). *Social change and crime rate trends: A routine activity approach* (Vol. 44). <https://doi.org/https://doi.org/10.2307/2094589>
- Cudis, C. (2021, March 17). *Public warned of increasing financial cybercrimes amid pandemic*. https://www.pna.gov.ph/articles/1133961?utm_source=&fbclid=IwY2xjawOBItZleHRuA2FibQlXMQBzcnRjBmFwcF9pZAEwAAEeeleRrOGTiZsdBdxYeUqYFvKJ7YtoaTB2XkHaem_RVYQoBhNPwZ_s4pRSN1elQ
- Dela Cruz, R. C. (2024). Smart blocks nearly 2B scam texts Jan to Oct 2024. <https://www.pna.gov.ph/articles/1239027>
- Dela Cruz, R. C. (2025). Filipinos receive over 6M text, 600K call scams in 2024. <https://www.pna.gov.ph/index.php/articles/1241859>
- der Veen, V. (2023, November). A BRIEF HISTORY OF FRAUD: NOTABLE CASES AND THE EVOLUTION OF SCAMS. <https://neetjevanderveen.com/f-4-a-brief-history-of-fraud-notable-cases-and-the-evolution-of-scams/?utm>
- Embudo, F. L. (2022). Most Links Sent Through Text Scams Are Gambling Sites - Cybersecurity Expert. <https://www.onenews.ph/articles/most-links-sent-through-text-scams-are-gambling-sites-cybersecurity-expert>
- Felson, M., & Clarke, R. V. G. (1998). *Opportunity makes the thief: Practical theory for crime prevention*. Home Office Policing; Reducing Crime Unit, Research, Development; Statistics Directorate.
- Gangwar, M., Mantri, S., Raveendra, S., & Shingenavar, K. (2024). *Telecom sim subscription frauds: Global policy trends, risk assessments, and recommendations* (tech. rep.). Indian School of Business.

- Ghimire, S. (2023). Timetrail: Unveiling financial fraud patterns through temporal correlation analysis. <http://arxiv.org/abs/2308.14215>
- GLOBE. (2024a). Globe Reiterates Public Warning Against SMS Spoofing, Calls for Increased Vigilance. <https://www.globe.com.ph/about-us/newsroom/corporate/vigilance-against-sms-spoofing>
- GLOBE. (2024b). Globe Sees 85% Decline in Scam, Spam SMS, Credits Strong Partnership with Banks Against Fraud. <https://www.globe.com.ph/about-us/newsroom/corporate/globe-partnership-with-banks-vs-fraud-sees-85-percent-decline-in-scam-spam-sms-blocked-as-of-q3-of-2023#gref>
- Gonzales, A. L. (2024). Kaspersky urges vigilance in PH as financial phishing rises. <https://www.pna.gov.ph/articles/1254230>
- Guiao, M. A. (2024). That Text From “GCash” or “BDO” With a Link Could be a Scam. https://www.spot.ph/newsfeatures/adulting/110218/how-to-protect-yourself-from-sms-spoofing-a5229-20241010?utm_source=copilot.com
- Hartig, F. (2024). *Dharma: Residual diagnostics for hierarchical (multi-level / mixed) regression models* [R package version 0.4.6]. <https://cran.r-project.org/package=DHARMA>
- Hilotin, J. (2022, January 18). *Philippines: 2,324% surge in hacking attacks*. <https://gulfnnews.com/business/philippines-2324-surge-in-hacking-attacks-1.1642489541309>
- Hox, J. (1998, January). *Multilevel modeling: When and why*. <https://doi.org/10.1007/978-3-642-72087-1\{-\}17>
- Kreft, I. G. G., & Yoon, B. (1994, April). Are multilevel techniques necessary? An attempt at demystification.
- Lenaerts-Bergmans, B. (2025, August). What is Spoofing? Spoofing Attacks Defined — CrowdStrike. <https://www.crowdstrike.com/en-us/cybersecurity-101/social-engineering/spoofing-attack/>
- Nahapetyan, A., Prasad, S., Childs, K., Oest, A., Ladwig, Y., Kapravelos, A., & Reaves, B. (2024). On sms phishing tactics and infrastructure. *Proceedings - IEEE Symposium on Security and Privacy*, 1–16. <https://doi.org/10.1109/SP54263.2024.00169>
- Nery, L. A., Catignas, R. D., & Tiam-Lee, T. J. (2025). KATOTOHANANQA: Evaluating truthfulness of large language models in Filipino. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2509.06065>
- Noble, J. (n.d.). What is an autoregressive model? [Accessed 2026]. *IBM*. <https://www.ibm.com/think/topics/autoregressive-model>
- Nyarko-Boateng, O., Nkrumah, B., Boafo, V., Amponsah, A., Tetteh, A., Oliver, K., Adekoya, A. F., Nti, I. K., Bawah, F. U., Songose, N. A., Samuel, B., Mensah, P. K., Nimbe, P., Adjepong, V. A., Agbedanu, R., & Weyori, B. A. (2022). A fraud prevention and secure cognitive sim card registration model. *Indian Journal of Science and Technology*, 15, 2562–2569. <https://doi.org/10.17485/IJST/v15i46.1658>
- Parrocha, A. (2022, October). Marcos signs ‘long overdue’ law curbing spread of text scams. <https://www.pna.gov.ph/articles/1185734>

- Parrocha, A. (2023, March). Marcos lauds nearly 100% decline in text scams due to SIM law. <https://www.pna.gov.ph/articles/1197505>
- PROTECT YOURSELF FROM FRAUD AND SCAMS* (tech. rep.). (n.d.). Bangko Sentral ng Pilipinas. https://www.bsp.gov.ph/Inclusive%5C%20Finance/Tools/Protect_yourself_from_Fraud_and_Scam.pdf
- Ramadan, R. A., Aboshosha, B. W., Alshudukhi, J. S., Alzahrani, A. J., El-Sayed, A., & Dessouky, M. M. (2021). Cybersecurity and countermeasures at the time of pandemic. <https://doi.org/10.1155/2021/6627264>
- Republic Act No. 11202: Mobile Number Portability Act (2018). https://lawphil.net/statutes/repacts/ra2019/ra_11202_2019.html
- Republic Act No. 11934: SIM Card Registration Act (2022). <https://www.officialgazette.gov.ph/2022/10/10/republic-act-no-11934/>
- Respicio & Co. (2025). Bank SMS Phishing in the Philippines: Chargeback, Complaints, and Legal Remedies. <https://www.lawyer-philippines.com/articles/bank-sms-phishing-in-the-philippines-chargeback-complaints-and-legal-remedies>
- Rosales, E. F. (2025). Gambling scams rise 76% as fraudsters exploit gaming surge. <https://www.philstar.com/headlines/2025/07/22/2459745/gambling-scams-rise-76-fraudsters-exploit-gaming-surge>
- Salman, M., Ikram, M., & Kaafar, M. A. (2024). Investigating evasive techniques in SMS spam Filtering: A Comparative Analysis of Machine Learning models. *IEEE Access*, 12, 24306–24324. <https://doi.org/10.1109/access.2024.3364671>
- Slater, P., & Hasson, F. (2024). Quantitative research designs, hierarchy of evidence and validity. *Journal of Psychiatric and Mental Health Nursing*, 32(3), 656–660. <https://doi.org/10.1111/jpm.13135>
- SyncTech – Android Apps. (n.d.). <https://www.synctech.com.au/>
- Thomas, D., & Zubkov, P. (2023, May). Quantitative research designs.
- Tumang, B. J. D. (2025). The sim registration legislation in enhancing mobile security against cyber threats.
- Wood, S. N. (2017, May). *Generalized additive models*. <https://doi.org/10.1201/9781315370279>
- Wood, S. N. (2025). *Mgcv: Mixed gam computation vehicle with automatic smoothness estimation* [R package documentation, version 1.9-4]. <https://cran.r-project.org/web/packages/mgcv/mgcv.pdf>
- Yang, A. Y. (2022). 'Mandatory SIM card registration puts privacy at risk, might not curb crime'. https://www.philstar.com/headlines/2022/02/04/2158543/mandatory-sim-card-registration-puts-privacy-risk-might-not-curb-crime?fbclid=IwY2xjawMc9WtleHRuA2FlbQlzMqABHI7AvvBkMCTIyhxnpmRQqeJLYG46u66MFQ-fVOnLJIA16dfgxRfudY1xW627_aem_x6jbBPQBpGEVHR2_DfVAyQ
- Yasar, K. (2023, April). smishing (SMS phishing). <https://www.techtarget.com/searchmobilecomputing/definition/SMiShing>

Zimba, A., Phiri, K., Kashale, C., & Phiri, M. (2025). Unveiling deception: a socio-economic analysis of smishing attacks on mobile money transaction users. *Humanities and Social Sciences Communications*, 12(1). <https://doi.org/10.1057/s41599-025-06141-8>